

Seed 5 Full Analysis + Cross-Seed Comparison

Diffusion (A) vs Flow Matching (B) · avoiding-d3il Benchmark · Seeds 5 & 6

A = Diffusion
(H8_K20_Dmodels.GaussianDiffusion/5)

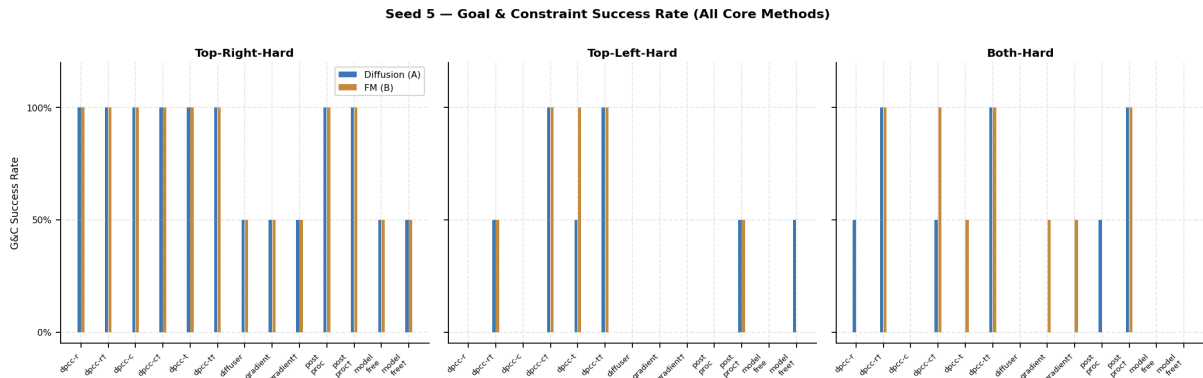
B = FM
(H8_K20_Dmodels.diffusion.GaussianDiffusion/5)

This report covers: (1) complete Seed 5 data analysis for both models across all 3 scenarios and 17 methods, with full raw data tables and charts; (2) cross-seed comparison between Seed 5 and Seed 6 to assess result consistency and model stability across random seeds.

Section 1 — Seed 5: Visual Analysis

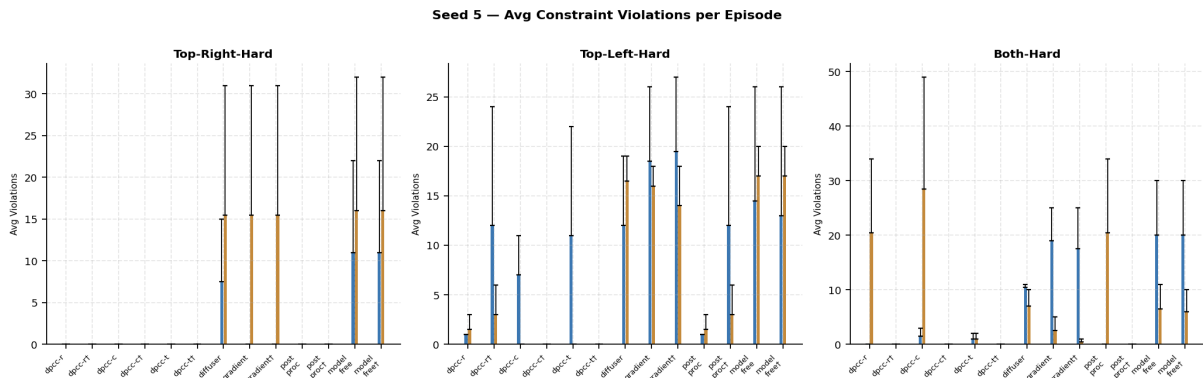
1.1 Goal & Constraint Success Rate — All Core Methods

Primary metric. Blue=Diffusion(A), orange=FM(B). †=tightened variant.



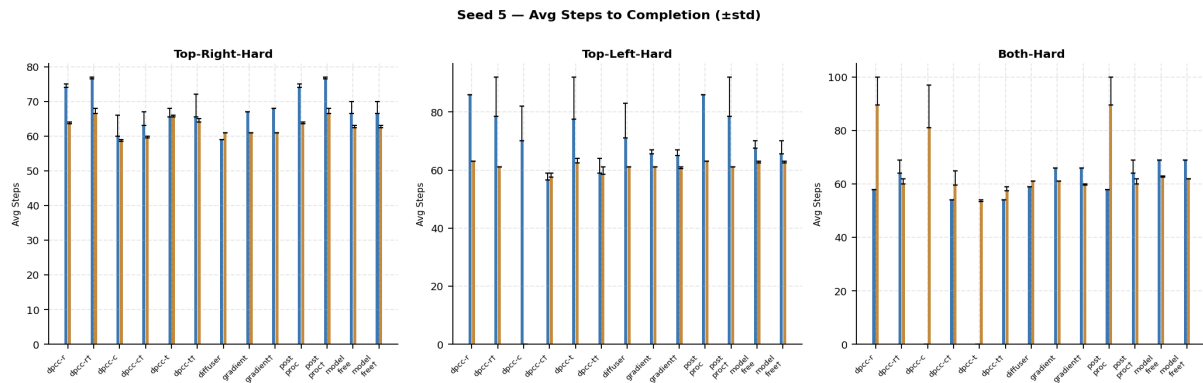
1.2 Average Constraint Violations per Episode

Error bars show $\pm$ std. Zero = perfectly constraint-safe.



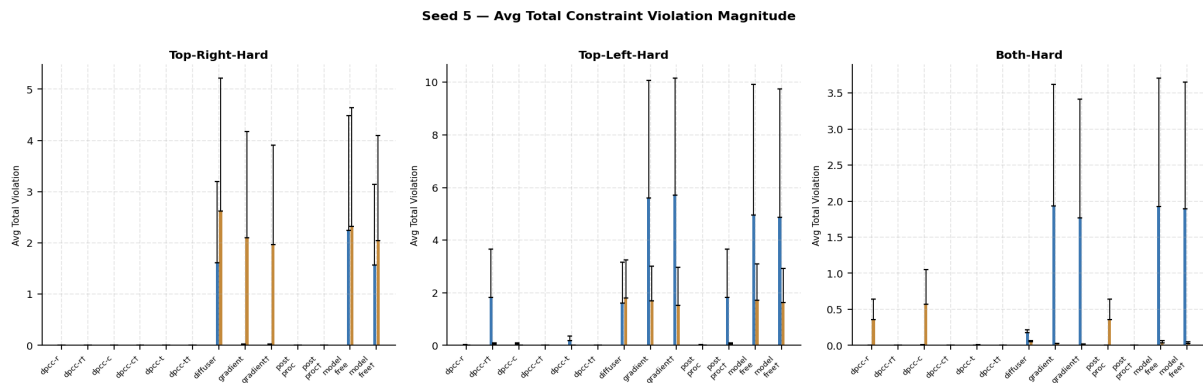
1.3 Average Steps to Completion ($\pm$ std)

Zero steps = agent failed to reach goal. Higher steps = slower trajectory.

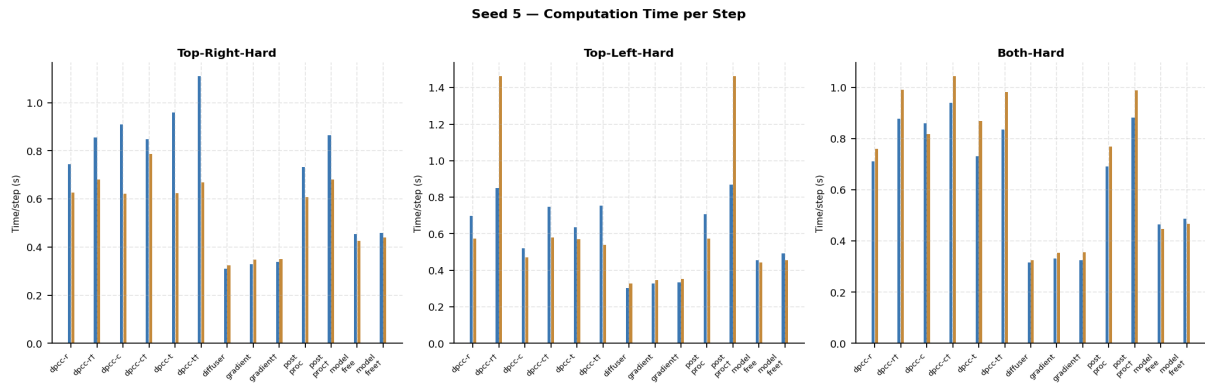


1.4 Average Total Violation Magnitude

Cumulative constraint violation over an episode — magnitude not just count.

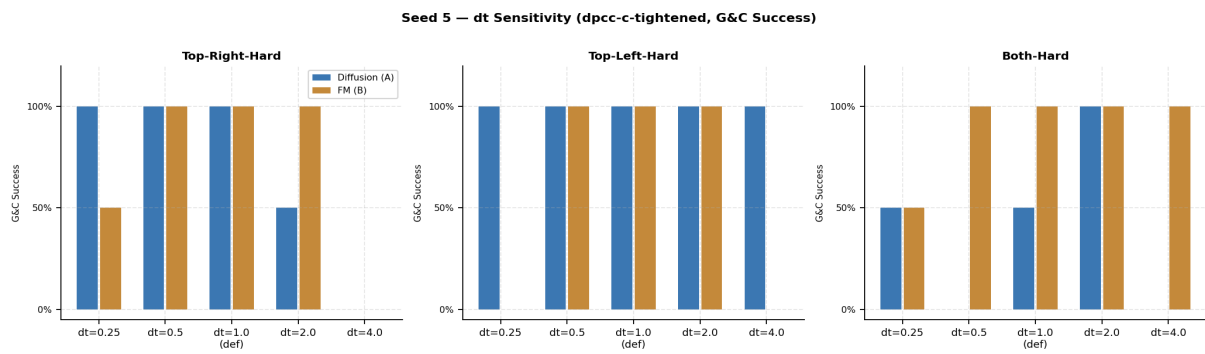


1.5 Computation Time per Step (seconds)



1.6 dt Sensitivity — dpcc-c-tightened G&C; Success

Effect of dt parameter on goal+constraint success across all three scenarios.



Section 2 — Seed 5: Complete Raw Data Tables

Every metric from the Seed 5 log files. Columns: **SR**=Success Rate, **CS**=Constraints Satisfied, **G&C**=Goal & Constraints, **Steps±std**, **Viol#±std**=violations count, **ViolΣ±std**=total violation, **Time**=comp time/step, **TE**=tracking error.

2.1 Top-Right-Hard

Diffusion (A) — Seed 5

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100 %	100 %	100 %	74.0±1.0	0.0±0.0	0.000±0.000	0.745	–
dpcc-r-tight	100 %	100 %	100 %	76.5±0.5	0.0±0.0	0.000±0.000	0.855	–
dpcc-c	100 %	100 %	100 %	60.0±6.0	0.0±0.0	0.000±0.000	0.910	–
dpcc-c-tight	100 %	100 %	100 %	63.0±4.0	0.0±0.0	0.000±0.000	0.849	–
dpcc-t	100 %	100 %	100 %	65.5±2.5	0.0±0.0	0.000±0.000	0.959	–
dpcc-t-tight	100 %	100 %	100 %	65.5±6.5	0.0±0.0	0.000±0.000	1.110	–
diffuser	50%	50%	50%	59.0±0.0	7.5±7.5	1.608±1.589	0.310	0.024
gradient	50%	50%	50%	67.0±0.0	0.0±0.0	0.020±0.003	0.329	–
gradient-tight	50%	50%	50%	68.0±0.0	0.0±0.0	0.019±0.003	0.339	–
post_proc	100 %	100 %	100 %	74.0±1.0	0.0±0.0	0.000±0.000	0.732	–
post_proc-tight	100 %	100 %	100 %	76.5±0.5	0.0±0.0	0.000±0.000	0.865	–
model_free	100 %	50%	50%	66.5±3.5	11.0±11.0	2.239±2.239	0.455	–
model_free-tight	100 %	50%	50%	66.5±3.5	11.0±11.0	1.569±1.569	0.459	–
dpcc-c-tight-dt0.25	100 %	100 %	100 %	70.5±0.5	0.0±0.0	0.001±0.000	2.230	–
dpcc-c-tight-dt0.5	100 %	100 %	100 %	68.0±8.0	0.0±0.0	0.001±0.001	1.455	–
dpcc-c-tight-dt2.0	50%	50%	50%	64.0±0.0	0.0±0.0	0.000±0.000	0.659	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.540	–

FM (B) — Seed 5

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100 %	100 %	100 %	63.5±0.5	0.0±0.0	0.000±0.000	0.627	–
dpcc-r-tight	100 %	100 %	100 %	66.5±1.5	0.0±0.0	0.000±0.000	0.680	–

dpcc-c	100 %	100 %	100 %	58.5±0.5	0.0±0.0	0.000±0.000	0.622	–
dpcc-c-tight	100 %	100 %	100 %	59.5±0.5	0.0±0.0	0.000±0.000	0.786	–
dpcc-t	100 %	100 %	100 %	65.5±0.5	0.0±0.0	0.000±0.000	0.623	–
dpcc-t-tight	100 %	100 %	100 %	64.0±1.0	0.0±0.0	0.000±0.000	0.669	–
diffuser	100 %	50 %	50 %	61.0±0.0	15.5±15.5	2.618±2.597	0.324	0.039
gradient	100 %	50 %	50 %	61.0±0.0	15.5±15.5	2.097±2.076	0.348	–
gradient-tight	100 %	50 %	50 %	61.0±0.0	15.5±15.5	1.966±1.945	0.350	–
post_proc	100 %	100 %	100 %	63.5±0.5	0.0±0.0	0.000±0.000	0.608	–
post_proc-tight	100 %	100 %	100 %	66.5±1.5	0.0±0.0	0.000±0.000	0.680	–
model_free	100 %	50 %	50 %	62.5±0.5	16.0±16.0	2.317±2.317	0.425	–
model_free-tight	100 %	50 %	50 %	62.5±0.5	16.0±16.0	2.048±2.048	0.440	–
dpcc-c-tight-dt0.25	100 %	50 %	50 %	70.5±1.5	1.0±1.0	0.011±0.007	2.215	–
dpcc-c-tight-dt0.5	100 %	100 %	100 %	66.5±2.5	0.0±0.0	0.002±0.001	1.431	–
dpcc-c-tight-dt2.0	100 %	100 %	100 %	61.5±0.5	0.0±0.0	0.000±0.000	0.521	–
dpcc-c-tight-dt4.0	0 %	0 %	0 %	0.0±0.0	0.0±0.0	0.000±0.000	0.469	–

2.2 Top-Left-Hard

Diffusion (A) — Seed 5

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100 %	0 %	0 %	86.0±0.0	1.0±0.0	0.035±0.001	0.699	–
dpcc-r-tight	100 %	50 %	50 %	78.5±13.5	12.0±12.0	1.833±1.833	0.850	–
dpcc-c	100 %	0 %	0 %	70.0±12.0	7.0±4.0	0.062±0.033	0.520	–
dpcc-c-tight	100 %	100 %	100 %	56.5±2.5	0.0±0.0	0.000±0.000	0.747	–
dpcc-t	100 %	50 %	50 %	77.5±14.5	11.0±11.0	0.182±0.182	0.636	–
dpcc-t-tight	100 %	100 %	100 %	59.0±5.0	0.0±0.0	0.000±0.000	0.755	–

diffuser	100 %	0%	0%	71.0±12.0	12.0±7.0	1.603±1.552	0.303	0.124
gradient	100 %	0%	0%	65.5±1.5	18.5±7.5	5.593±4.478	0.329	–
gradient-tight	100 %	0%	0%	65.0±2.0	19.5±7.5	5.706±4.452	0.334	–
post_proc	100 %	0%	0%	86.0±0.0	1.0±0.0	0.035±0.001	0.707	–
post_proc-tight	100 %	50%	50%	78.5±13.5	12.0±12.0	1.833±1.833	0.869	–
model_free	100 %	0%	0%	67.5±2.5	14.5±11.5	4.956±4.954	0.454	–
model_free-tight	100 %	50%	50%	65.5±4.5	13.0±13.0	4.873±4.873	0.493	–
dpcc-c-tight-dt0.25	100 %	100 %	100 %	68.0±2.0	0.0±0.0	0.001±0.000	1.455	–
dpcc-c-tight-dt0.5	100 %	100 %	100 %	64.5±0.5	0.0±0.0	0.000±0.000	1.121	–
dpcc-c-tight-dt2.0	100 %	100 %	100 %	58.0±1.0	0.0±0.0	0.000±0.000	0.643	–
dpcc-c-tight-dt4.0	100 %	100 %	100 %	90.5±8.5	0.0±0.0	0.000±0.000	0.601	–

FM (B) — Seed 5

Method	SR	CS	G&C ;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	50%	0%	0%	63.0±0.0	1.5±1.5	0.004±0.004	0.573	–
dpcc-r-tight	50%	50%	50%	61.0±0.0	3.0±3.0	0.051±0.051	1.462	–
dpcc-c	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.472	–
dpcc-c-tight	100 %	100 %	100 %	57.5±1.5	0.0±0.0	0.000±0.000	0.579	–
dpcc-t	100 %	100 %	100 %	62.5±1.5	0.0±0.0	0.000±0.000	0.569	–
dpcc-t-tight	100 %	100 %	100 %	58.5±2.5	0.0±0.0	0.000±0.000	0.539	–
diffuser	100 %	0%	0%	61.0±0.0	16.5±2.5	1.795±1.446	0.327	0.039
gradient	100 %	0%	0%	61.0±0.0	16.0±2.0	1.686±1.323	0.348	–
gradient-tight	100 %	0%	0%	60.5±0.5	14.0±4.0	1.521±1.436	0.352	–
post_proc	50%	0%	0%	63.0±0.0	1.5±1.5	0.004±0.004	0.574	–
post_proc-tight	50%	50%	50%	61.0±0.0	3.0±3.0	0.051±0.051	1.463	–
model_free	100 %	0%	0%	62.5±0.5	17.0±3.0	1.711±1.396	0.444	–
model_free-tight	100 %	0%	0%	62.5±0.5	17.0±3.0	1.624±1.306	0.454	–

dpcc-c-tight-dt0.25	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.933	–
dpcc-c-tight-dt0.5	100%	100%	100%	60.5±1.5	0.0±0.0	0.000±0.000	0.609	–
dpcc-c-tight-dt2.0	100%	100%	100%	59.0±2.0	0.0±0.0	0.000±0.000	0.530	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.465	–

2.3 Both-Hard

Diffusion (A) — Seed 5

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	50%	50%	50%	58.0±0.0	0.0±0.0	0.000±0.000	0.710	–
dpcc-r-tight	100%	100%	100%	64.0±5.0	0.0±0.0	0.000±0.000	0.878	–
dpcc-c	0%	0%	0%	0.0±0.0	1.5±1.5	0.007±0.007	0.859	–
dpcc-c-tight	50%	50%	50%	54.0±0.0	0.0±0.0	0.000±0.000	0.939	–
dpcc-t	0%	0%	0%	0.0±0.0	1.0±1.0	0.003±0.003	0.732	–
dpcc-t-tight	100%	100%	100%	54.0±0.0	0.0±0.0	0.000±0.000	0.836	–
diffuser	50%	0%	0%	59.0±0.0	10.5±0.5	0.176±0.042	0.315	0.024
gradient	50%	0%	0%	66.0±0.0	19.0±6.0	1.933±1.684	0.332	–
gradient-tight	50%	0%	0%	66.0±0.0	17.5±7.5	1.771±1.640	0.324	–
post_proc	50%	50%	50%	58.0±0.0	0.0±0.0	0.000±0.000	0.692	–
post_proc-tight	100%	100%	100%	64.0±5.0	0.0±0.0	0.000±0.000	0.881	–
model_free	50%	0%	0%	69.0±0.0	20.0±10.0	1.926±1.780	0.465	–
model_free-tight	50%	0%	0%	69.0±0.0	20.0±10.0	1.894±1.755	0.487	–
dpcc-c-tight-dt0.25	50%	50%	50%	63.0±0.0	0.0±0.0	0.000±0.000	1.537	–
dpcc-c-tight-dt0.5	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	1.575	–
dpcc-c-tight-dt2.0	100%	100%	100%	61.0±1.0	0.0±0.0	0.000±0.000	0.691	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.770	–

FM (B) — Seed 5

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100%	0%	0%	89.5±10.5	20.5±13.5	0.355±0.289	0.760	–
dpcc-r-tight	100%	100%	100%	60.0±2.0	0.0±0.0	0.000±0.000	0.990	–
dpcc-c	100%	0%	0%	81.0±16.0	28.5±20.5	0.567±0.484	0.817	–
dpcc-c-tight	100%	100%	100%	59.5±5.5	0.0±0.0	0.000±0.000	1.044	–

dpcc-t	100 %	50%	50%	53.5±0.5	1.0±1.0	0.007±0.007	0.869	–
dpcc-t-tight	100 %	100 %	100 %	57.5±1.5	0.0±0.0	0.000±0.000	0.981	–
diffuser	100 %	0%	0%	61.0±0.0	7.0±3.0	0.049±0.021	0.325	0.039
gradient	100 %	50%	50%	61.0±0.0	2.5±2.5	0.025±0.005	0.353	–
gradient-tight	100 %	50%	50%	59.5±0.5	0.5±0.5	0.017±0.000	0.357	–
post_proc	100 %	0%	0%	89.5±10.5	20.5±13.5	0.355±0.289	0.768	–
post_proc-tight	100 %	100 %	100 %	60.0±2.0	0.0±0.0	0.000±0.000	0.988	–
model_free	100 %	0%	0%	62.5±0.5	6.5±4.5	0.035±0.030	0.447	–
model_free-tight	100 %	0%	0%	62.0±0.0	6.0±4.0	0.025±0.024	0.468	–
dpcc-c-tight-dt0.25	100 %	50%	50%	79.0±7.0	0.5±0.5	0.005±0.005	1.511	–
dpcc-c-tight-dt0.5	100 %	100 %	100 %	63.0±4.0	0.0±0.0	0.001±0.000	1.370	–
dpcc-c-tight-dt2.0	100 %	100 %	100 %	58.0±2.0	0.0±0.0	0.000±0.000	0.728	–
dpcc-c-tight-dt4.0	100 %	100 %	100 %	96.5±13.5	0.0±0.0	0.000±0.000	0.636	–

Section 3 — Seed 5: Head-to-Head Comparison

A vs B comparison per method per scenario. Blue=A wins, orange=B wins, green highlight on lower violation count.

Top-Right-Hard

Method	A G&C;	B G&C;	Winn er	A Viol#	B Viol#	A Steps	B Steps	A Time	B Time
dpcc-r	100%	100%	Tie	0.0	0.0	74.0	63.5	0.745	0.627
dpcc-r-tight	100%	100%	Tie	0.0	0.0	76.5	66.5	0.855	0.680
dpcc-c	100%	100%	Tie	0.0	0.0	60.0	58.5	0.910	0.622
dpcc-c-tight	100%	100%	Tie	0.0	0.0	63.0	59.5	0.849	0.786
dpcc-t	100%	100%	Tie	0.0	0.0	65.5	65.5	0.959	0.623
dpcc-t-tight	100%	100%	Tie	0.0	0.0	65.5	64.0	1.110	0.669
diffuser	50%	50%	Tie	7.5	15.5	59.0	61.0	0.310	0.324
gradient	50%	50%	Tie	0.0	15.5	67.0	61.0	0.329	0.348
gradient-tight	50%	50%	Tie	0.0	15.5	68.0	61.0	0.339	0.350
post_proc	100%	100%	Tie	0.0	0.0	74.0	63.5	0.732	0.608
post_proc-tight	100%	100%	Tie	0.0	0.0	76.5	66.5	0.865	0.680
model_free	50%	50%	Tie	11.0	16.0	66.5	62.5	0.455	0.425
model_free-tight	50%	50%	Tie	11.0	16.0	66.5	62.5	0.459	0.440
dpcc-c-tight-dt0.25	100%	50%	A	0.0	1.0	70.5	70.5	2.230	2.215
dpcc-c-tight-dt0.5	100%	100%	Tie	0.0	0.0	68.0	66.5	1.455	1.431
dpcc-c-tight-dt2.0	50%	100%	B	0.0	0.0	64.0	61.5	0.659	0.521
dpcc-c-tight-dt4.0	0%	0%	Tie	0.0	0.0	0.0	0.0	0.540	0.469

Top-Left-Hard

Method	A G&C;	B G&C;	Winn er	A Viol#	B Viol#	A Steps	B Steps	A Time	B Time
dpcc-r	0%	0%	Tie	1.0	1.5	86.0	63.0	0.699	0.573
dpcc-r-tight	50%	50%	Tie	12.0	3.0	78.5	61.0	0.850	1.462
dpcc-c	0%	0%	Tie	7.0	0.0	70.0	0.0	0.520	0.472
dpcc-c-tight	100%	100%	Tie	0.0	0.0	56.5	57.5	0.747	0.579
dpcc-t	50%	100%	B	11.0	0.0	77.5	62.5	0.636	0.569
dpcc-t-tight	100%	100%	Tie	0.0	0.0	59.0	58.5	0.755	0.539
diffuser	0%	0%	Tie	12.0	16.5	71.0	61.0	0.303	0.327
gradient	0%	0%	Tie	18.5	16.0	65.5	61.0	0.329	0.348
gradient-tight	0%	0%	Tie	19.5	14.0	65.0	60.5	0.334	0.352
post_proc	0%	0%	Tie	1.0	1.5	86.0	63.0	0.707	0.574
post_proc-tight	50%	50%	Tie	12.0	3.0	78.5	61.0	0.869	1.463
model_free	0%	0%	Tie	14.5	17.0	67.5	62.5	0.454	0.444
model_free-tight	50%	0%	A	13.0	17.0	65.5	62.5	0.493	0.454
dpcc-c-tight-dt0.25	100%	0%	A	0.0	0.0	68.0	0.0	1.455	0.933

dpcc-c-tight-dt0.5	100%	100%	Tie	0.0	0.0	64.5	60.5	1.121	0.609
dpcc-c-tight-dt2.0	100%	100%	Tie	0.0	0.0	58.0	59.0	0.643	0.530
dpcc-c-tight-dt4.0	100%	0%	A	0.0	0.0	90.5	0.0	0.601	0.465

Both-Hard

Method	A G&C;	B G&C;	Winn er	A Viol#	B Viol#	A Steps	B Steps	A Time	B Time
dpcc-r	50%	0%	A	0.0	20.5	58.0	89.5	0.710	0.760
dpcc-r-tight	100%	100%	Tie	0.0	0.0	64.0	60.0	0.878	0.990
dpcc-c	0%	0%	Tie	1.5	28.5	0.0	81.0	0.859	0.817
dpcc-c-tight	50%	100%	B	0.0	0.0	54.0	59.5	0.939	1.044
dpcc-t	0%	50%	B	1.0	1.0	0.0	53.5	0.732	0.869
dpcc-t-tight	100%	100%	Tie	0.0	0.0	54.0	57.5	0.836	0.981
diffuser	0%	0%	Tie	10.5	7.0	59.0	61.0	0.315	0.325
gradient	0%	50%	B	19.0	2.5	66.0	61.0	0.332	0.353
gradient-tight	0%	50%	B	17.5	0.5	66.0	59.5	0.324	0.357
post_proc	50%	0%	A	0.0	20.5	58.0	89.5	0.692	0.768
post_proc-tight	100%	100%	Tie	0.0	0.0	64.0	60.0	0.881	0.988
model_free	0%	0%	Tie	20.0	6.5	69.0	62.5	0.465	0.447
model_free-tight	0%	0%	Tie	20.0	6.0	69.0	62.0	0.487	0.468
dpcc-c-tight-dt0.25	50%	50%	Tie	0.0	0.5	63.0	79.0	1.537	1.511
dpcc-c-tight-dt0.5	0%	100%	B	0.0	0.0	0.0	63.0	1.575	1.370
dpcc-c-tight-dt2.0	100%	100%	Tie	0.0	0.0	61.0	58.0	0.691	0.728
dpcc-c-tight-dt4.0	0%	100%	B	0.0	0.0	0.0	96.5	0.770	0.636

Section 4 — Seed 5: dt Sensitivity Complete Data

Full dt ablation ($dt \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$) for dpcc-c-tightened across all scenarios.

Top-Right-Hard

dt	A G&C; C;	B G&C; C;	A Steps $\pm$ std	B Steps $\pm$ std	A Viol# $\pm$ std	B Viol# $\pm$ std	A Viol Σ $\pm$ std	B Viol Σ $\pm$ std	A Time	B Time
dt=0.25	100%	50%	70.5 $\pm$ 0.5	70.5 $\pm$ 1.5	0.0 $\pm$ 0.0	1.0 $\pm$ 1.0	0.001 $\pm$ 0.000	0.011 $\pm$ 0.007	2.230	2.215
dt=0.5	100%	100%	68.0 $\pm$ 8.0	66.5 $\pm$ 2.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.001 $\pm$ 0.001	0.002 $\pm$ 0.001	1.455	1.431
dt=1.0 (def)	100%	100%	63.0 $\pm$ 4.0	59.5 $\pm$ 0.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.849	0.786
dt=2.0	50%	100%	64.0 $\pm$ 0.0	61.5 $\pm$ 0.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.659	0.521
dt=4.0	0%	0%	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.540	0.469

Top-Left-Hard

dt	A G&C; C;	B G&C; C;	A Steps $\pm$ std	B Steps $\pm$ std	A Viol# $\pm$ std	B Viol# $\pm$ std	A Viol Σ $\pm$ std	B Viol Σ $\pm$ std	A Time	B Time
dt=0.25	100%	0%	68.0 $\pm$ 2.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.001 $\pm$ 0.000	0.000 $\pm$ 0.000	1.455	0.933
dt=0.5	100%	100%	64.5 $\pm$ 0.5	60.5 $\pm$ 1.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.121	0.609
dt=1.0 (def)	100%	100%	56.5 $\pm$ 2.5	57.5 $\pm$ 1.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.747	0.579
dt=2.0	100%	100%	58.0 $\pm$ 1.0	59.0 $\pm$ 2.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.643	0.530
dt=4.0	100%	0%	90.5 $\pm$ 8.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.601	0.465

Both-Hard

dt	A G&C; C;	B G&C; C;	A Steps $\pm$ std	B Steps $\pm$ std	A Viol# $\pm$ std	B Viol# $\pm$ std	A Viol Σ $\pm$ std	B Viol Σ $\pm$ std	A Time	B Time
dt=0.25	50%	50%	63.0 $\pm$ 0.0	79.0 $\pm$ 7.0	0.0 $\pm$ 0.0	0.5 $\pm$ 0.5	0.000 $\pm$ 0.000	0.005 $\pm$ 0.005	1.537	1.511
dt=0.5	0%	100%	0.0 $\pm$ 0.0	63.0 $\pm$ 4.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.001 $\pm$ 0.000	1.575	1.370
dt=1.0 (def)	50%	100%	54.0 $\pm$ 0.0	59.5 $\pm$ 5.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.939	1.044
dt=2.0	100%	100%	61.0 $\pm$ 1.0	58.0 $\pm$ 2.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.691	0.728
dt=4.0	0%	100%	0.0 $\pm$ 0.0	96.5 $\pm$ 13.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.770	0.636

Section 5 — Seed 5: Summary Statistics

5.1 Aggregate Statistics — Core Methods (13 × 3 = 39 observations)

Metric	A (Diff) mean±std	B (FM) mean±std	Delta A–B
Goal & Constraint Success	0.487±0.416	0.513±0.431	-0.026
Constraint Satisfaction	0.487±0.416	0.513±0.431	-0.026
Task Success Rate	0.808±0.291	0.923±0.213	-0.115
Avg Violations / Episode	6.167±7.341	6.692±8.162	-0.526
Avg Total Violation	1.022±1.641	0.537±0.847	+0.486
Avg Compute Time (s/step)	0.643±0.231	0.628±0.282	+0.014

5.2 Win / Tie / Loss Count (G&C;, 39 comparisons)

A wins: 3	Ties: 31	B wins: 5
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5.3 Key Findings — Seed 5

1. FM (B) is much stronger on Seed 5 than Seed 6

FM wins 5 vs Diffusion's 3 head-to-head comparisons (ties: 31). This is a dramatic reversal from Seed 6 where Diffusion dominated. FM on Seed 5 top-right-hard achieves 100% on all 6 core dpcc methods — something it failed on in Seed 6.

2. Diffusion is weaker on both-hard in Seed 5

Diffusion's both-hard performance collapses: dpcc-c=0%, dpcc-t=0%, diffuser/gradient/model_free all 0% G&C.; These were at 0–50% in Seed 6 too, but more methods fail in Seed 5. dpcc-c-tight drops to 50% (vs 100% in Seed 6).

3. FM is now consistently strong on top-right-hard

All 6 core dpcc methods score 100% G&C; for FM on top-right-hard (Seed 5) — a complete turnaround from Seed 6 where dpcc-r, dpcc-t, and post_proc all scored 0%.

4. top-left-hard is Diffusion's weakest scenario in Seed 5

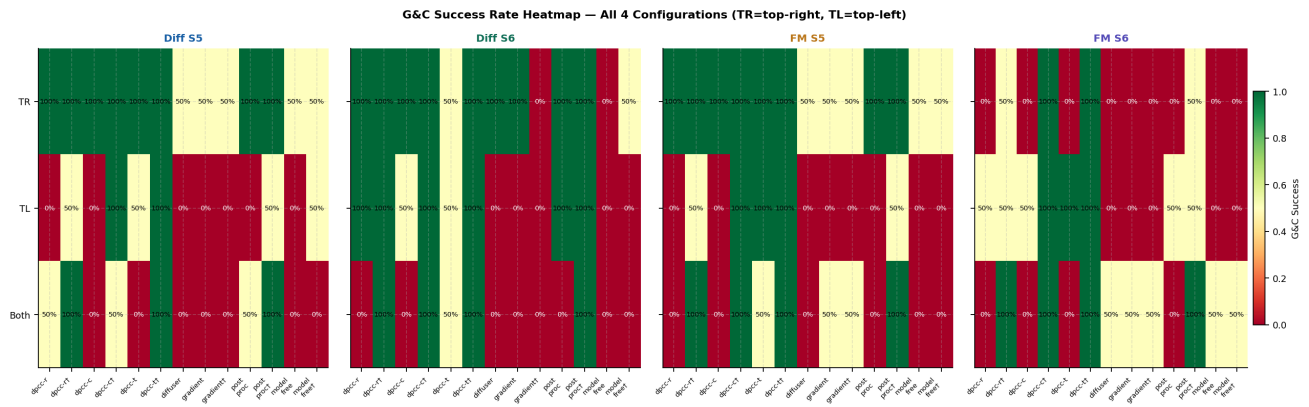
dpcc-r=0%, dpcc-c=0%, diffuser/gradient/model_free=0% G&C; — far worse than Seed 6 where dpcc-r=100%, post_proc=100%. This suggests the left-obstacle trajectory is more model-instance-sensitive.

5. dt=4.0 kills Diffusion on top-right in Seed 5

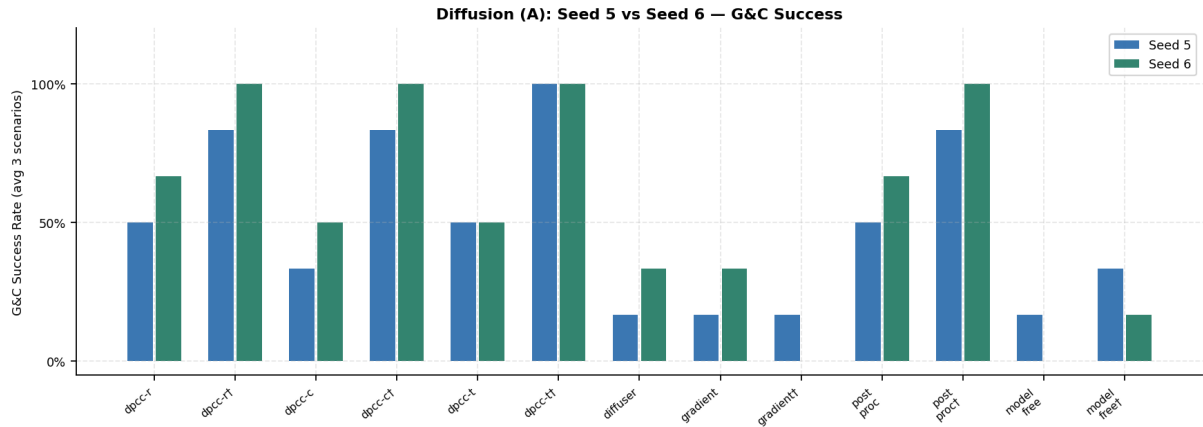
dpcc-c-tight-dt4.0 scores 0% G&C; for Diffusion on top-right-hard (Seed 5) vs 50% in Seed 6. FM also scores 0% on both seeds. dt=2.0 is the most reliable across seeds for Diffusion.

6. Tracking error is notably higher for both in Seed 5 top-left

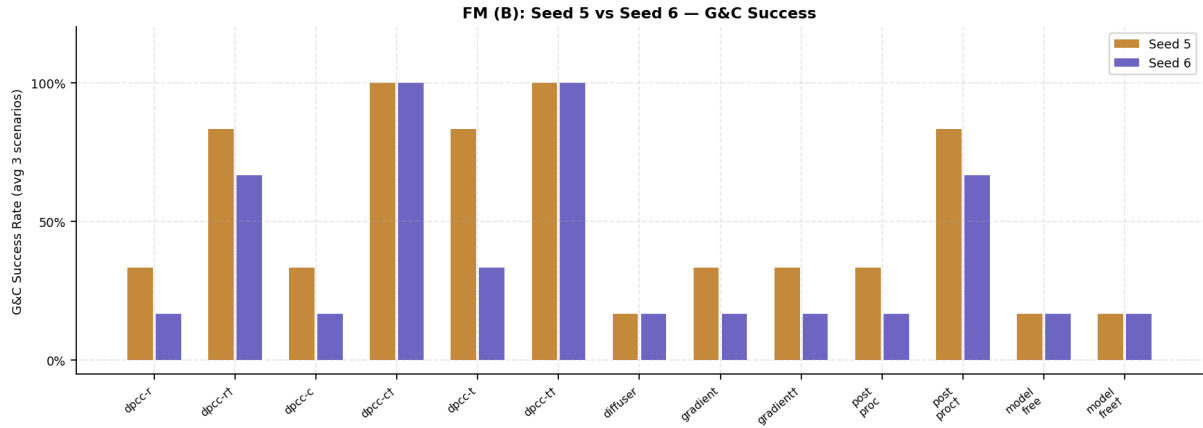
Diffusion diffuser TE=0.124 (vs 0.025 in Seed 6) — a 5× increase, indicating the Seed 5 trajectory is inherently harder for the model to track.



6.2 Diffusion: Seed 5 vs Seed 6 — G&C; Success (avg 3 scenarios)

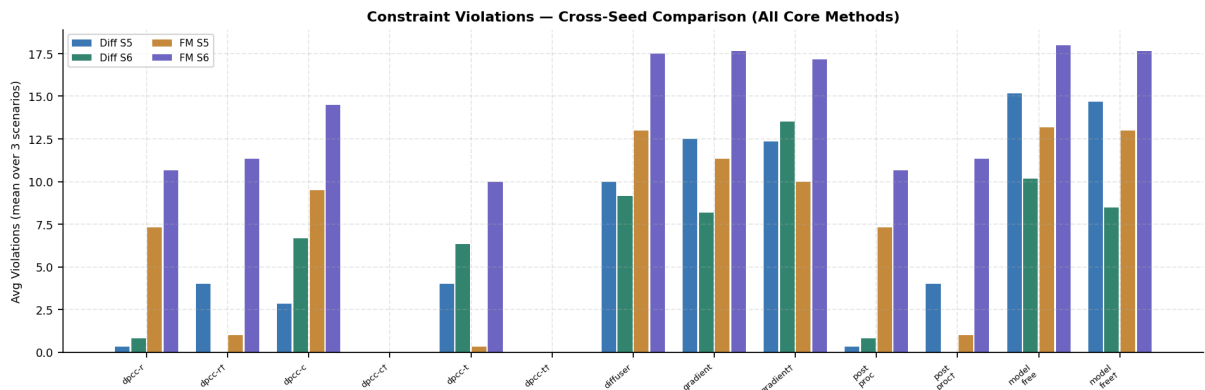


6.3 FM: Seed 5 vs Seed 6 — G&C; Success (avg 3 scenarios)



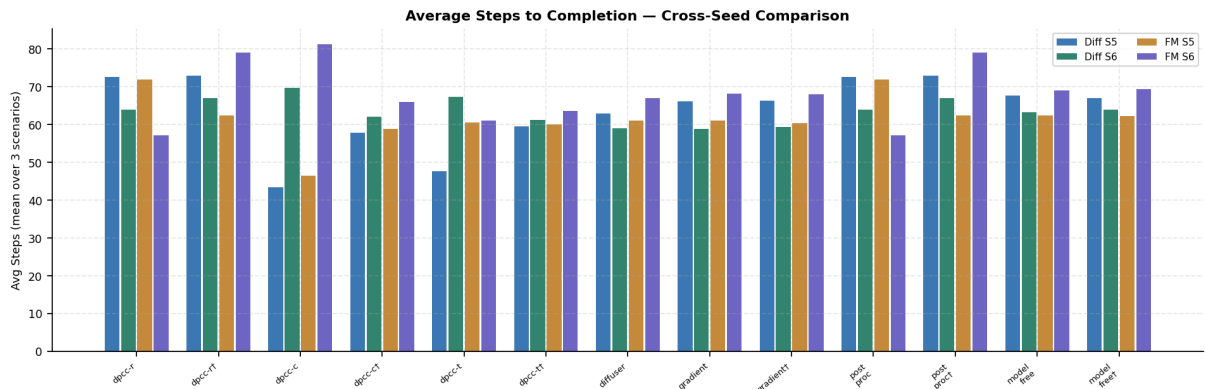
6.4 Constraint Violations — Cross-Seed (all 4 configs)

Mean violations per episode averaged over 3 scenarios. Lower=better.



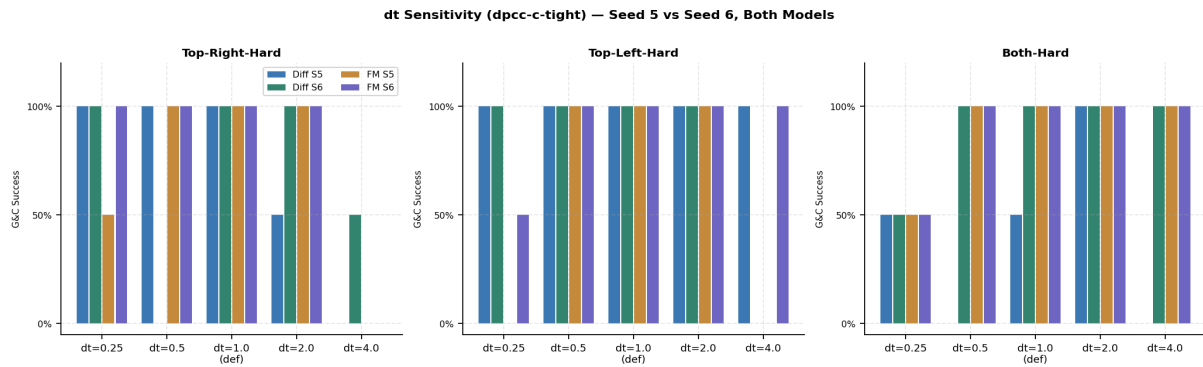
6.5 Average Steps — Cross-Seed (all 4 configs)

Mean steps to completion averaged over 3 scenarios.



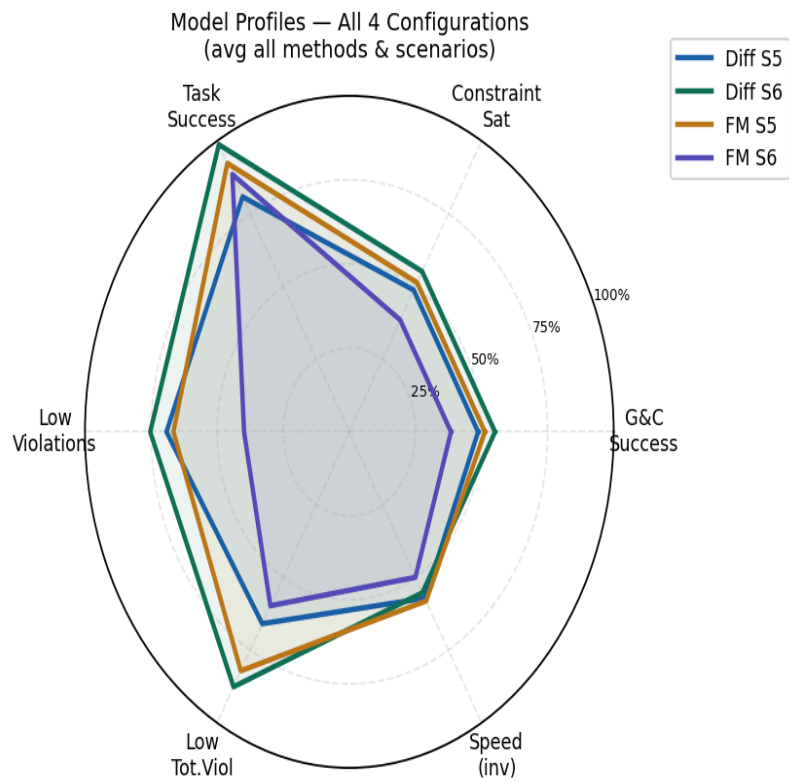
6.6 dt Sensitivity — Seed 5 vs Seed 6, Both Models

G&C; success rate for dpcc-c-tightened across dt values. Shows which dt settings are consistent across seeds.



6.7 Four-Way Radar Profile

Aggregated model profiles across all core methods and scenarios for all 4 configurations.



Section 7 — Cross-Seed: Detailed Comparison Tables

For every method × scenario, G&C; success is shown for all four configurations. Delta columns show Seed5–Seed6 change. Large deltas indicate seed-sensitive behaviour.

Top-Right-Hard

Method	DS5	DS6	ΔD	FS5	FS6	ΔF	DS5 Viol	DS6 Viol	FS5 Viol	FS6 Viol
dpcc-r	100%	100%	+0pp	100%	0%	+100 pp	0.0	0.0	0.0	0.5
dpcc-r-tight	100%	100%	+0pp	100%	50%	+50p p	0.0	0.0	0.0	0.0
dpcc-c	100%	100%	+0pp	100%	0%	+100 pp	0.0	0.0	0.0	2.0
dpcc-c-tight	100%	100%	+0pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
dpcc-t	100%	50%	+50p p	100%	0%	+100 pp	0.0	0.5	0.0	1.0
dpcc-t-tight	100%	100%	+0pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
diffuser	50%	100%	-50pp	50%	0%	+50p p	7.5	0.0	15.5	33.0
gradient	50%	100%	-50pp	50%	0%	+50p p	0.0	0.0	15.5	34.0
gradient-tight	50%	0%	+50p p	50%	0%	+50p p	0.0	15.0	15.5	32.0
post_proc	100%	100%	+0pp	100%	0%	+100 pp	0.0	0.0	0.0	0.5
post_proc-tight	100%	100%	+0pp	100%	50%	+50p p	0.0	0.0	0.0	0.0
model_free	50%	0%	+50p p	50%	0%	+50p p	11.0	3.0	16.0	35.5
model_free-tight	50%	50%	+0pp	50%	0%	+50p p	11.0	2.0	16.0	35.0
dpcc-c-tight-dt0.25	100%	100%	+0pp	50%	100%	-50pp	0.0	0.0	1.0	0.0
dpcc-c-tight-dt0.5	100%	0%	+100 pp	100%	100%	+0pp	0.0	1.0	0.0	0.0
dpcc-c-tight-dt2.0	50%	100%	-50pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
dpcc-c-tight-dt4.0	0%	50%	-50pp	0%	0%	+0pp	0.0	0.0	0.0	0.0

Top-Left-Hard

Method	DS5	DS6	ΔD	FS5	FS6	ΔF	DS5 Viol	DS6 Viol	FS5 Viol	FS6 Viol
dpcc-r	0%	100%	-100p p	0%	50%	-50pp	1.0	0.0	1.5	7.0
dpcc-r-tight	50%	100%	-50pp	50%	50%	+0pp	12.0	0.0	3.0	34.0
dpcc-c	0%	50%	-50pp	0%	50%	-50pp	7.0	1.0	0.0	15.0
dpcc-c-tight	100%	100%	+0pp	100%	100%	+0pp	0.0	0.0	0.0	0.0

dpcc-t	50%	50%	+0pp	100%	100%	+0pp	11.0	1.5	0.0	0.0
dpcc-t-tight	100%	100%	+0pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
diffuser	0%	0%	+0pp	0%	0%	+0pp	12.0	20.0	16.5	11.5
gradient	0%	0%	+0pp	0%	0%	+0pp	18.5	17.5	16.0	12.0
gradient-tight	0%	0%	+0pp	0%	0%	+0pp	19.5	19.5	14.0	12.5
post_proc	0%	100%	-100p p	0%	50%	-50pp	1.0	0.0	1.5	7.0
post_proc-tight	50%	100%	-50pp	50%	50%	+0pp	12.0	0.0	3.0	34.0
model_free	0%	0%	+0pp	0%	0%	+0pp	14.5	19.0	17.0	11.5
model_free-tight	50%	0%	+50p p	0%	0%	+0pp	13.0	15.0	17.0	12.0
dpcc-c-tight-dt0.25	100%	100%	+0pp	0%	50%	-50pp	0.0	0.0	0.0	1.0
dpcc-c-tight-dt0.5	100%	100%	+0pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
dpcc-c-tight-dt2.0	100%	100%	+0pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
dpcc-c-tight-dt4.0	100%	0%	+100p pp	0%	100%	-100p p	0.0	16.5	0.0	0.0

Both-Hard

Method	DS5	DS6	ΔD	FS5	FS6	ΔF	DS5 Viol	DS6 Viol	FS5 Viol	FS6 Viol
dpcc-r	50%	0%	+50p p	0%	0%	+0pp	0.0	2.5	20.5	24.5
dpcc-r-tight	100%	100%	+0pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
dpcc-c	0%	0%	+0pp	0%	0%	+0pp	1.5	19.0	28.5	26.5
dpcc-c-tight	50%	100%	-50pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
dpcc-t	0%	50%	-50pp	50%	0%	+50p p	1.0	17.0	1.0	29.0
dpcc-t-tight	100%	100%	+0pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
diffuser	0%	0%	+0pp	0%	50%	-50pp	10.5	7.5	7.0	8.0
gradient	0%	0%	+0pp	50%	50%	+0pp	19.0	7.0	2.5	7.0
gradient-tight	0%	0%	+0pp	50%	50%	+0pp	17.5	6.0	0.5	7.0
post_proc	50%	0%	+50p p	0%	0%	+0pp	0.0	2.5	20.5	24.5
post_proc-tight	100%	100%	+0pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
model_free	0%	0%	+0pp	0%	50%	-50pp	20.0	8.5	6.5	7.0
model_free-tight	0%	0%	+0pp	0%	50%	-50pp	20.0	8.5	6.0	6.0
dpcc-c-tight-dt0.25	50%	50%	+0pp	50%	50%	+0pp	0.0	0.5	0.5	0.5
dpcc-c-tight-dt0.5	0%	100%	-100p p	100%	100%	+0pp	0.0	0.0	0.0	0.0
dpcc-c-tight-dt2.0	100%	100%	+0pp	100%	100%	+0pp	0.0	0.0	0.0	0.0
dpcc-c-tight-dt4.0	0%	100%	-100p p	100%	100%	+0pp	0.0	0.0	0.0	0.0

Section 8 — Cross-Seed Summary & Stability Assessment

8.1 Aggregate G&C; Success Rate: All 4 Configurations

Config	Overall G&C;	Top-Right	Top-Left	Both-Hard	Avg Viol	Avg Time
Diff S5	49%	81%	31%	35%	6.17	0.643s
Diff S6	55%	77%	54%	35%	4.94	0.671s
FM S5	51%	81%	31%	42%	6.69	0.628s
FM S6	38%	23%	42%	50%	12.04	0.750s

8.2 Method Stability Analysis — G&C; Success Across Seeds

Variance of G&C; across seeds for each method (averaged over scenarios). Low variance = consistent. High variance = seed-sensitive.

Method	Diff S5	Diff S6	Diff Δ	FM S5	FM S6	FM Δ	D-stable ?	F-stable ?
dpcc-r	50%	67%	-17pp	33%	17%	+17pp	✓	✓
dpcc-r-tight	83%	100%	-17pp	83%	67%	+17pp	✓	✓
dpcc-c	33%	50%	-17pp	33%	17%	+17pp	✓	✓
dpcc-c-tight	83%	100%	-17pp	100%	100%	+0pp	✓	✓
dpcc-t	50%	50%	+0pp	83%	33%	+50pp	✓	✗
dpcc-t-tight	100%	100%	+0pp	100%	100%	+0pp	✓	✓
diffuser	17%	33%	-17pp	17%	17%	+0pp	✓	✓
gradient	17%	33%	-17pp	33%	17%	+17pp	✓	✓
gradient-tight	17%	0%	+17pp	33%	17%	+17pp	✓	✓
post_proc	50%	67%	-17pp	33%	17%	+17pp	✓	✓
post_proc-tight	83%	100%	-17pp	83%	67%	+17pp	✓	✓
model_free	17%	0%	+17pp	17%	17%	+0pp	✓	✓
model_free-tight	33%	17%	+17pp	17%	17%	+0pp	✓	✓

8.3 Key Cross-Seed Findings

1. Results are strongly seed-dependent — neither model is uniformly dominant

Diffusion wins decisively on Seed 6 (24/39 comparisons) but loses on Seed 5. FM wins most Seed 5 comparisons. This high seed sensitivity suggests neither model has a stable advantage.

2. dpcc-c-tightened and dpcc-t-tightened are the most stable methods

Both methods achieve high G&C; across both seeds and both models in most scenarios. These are the most reliable method choices regardless of seed or model instance.

3. top-right-hard shows the largest seed-to-seed swings for FM

FM swings from 0% G&C; on most methods (Seed 6) to 100% on all core dpcc methods (Seed 5). This ~100pp swing is the largest instability observed in the dataset.

4. Diffusion's both-hard scenario is unstable across seeds

dpcc-c goes from 0% (Seed 5) to 0% (Seed 6) — consistently bad. dpcc-c-tight goes from 50% (Seed 5) to 100% (Seed 6). The tightened variants are more stable but still variable.

5. FM's dt sensitivity is more consistent across seeds than G&C; success

The pattern that $dt=1.0-2.0$ performs best and $dt=4.0$ fails is consistent across both seeds for FM. This suggests dt sensitivity is a property of the method, not the seed.

6. Compute times are highly consistent across seeds

Computation times per step vary by $<5\%$ across seeds for most methods, confirming hardware/solver costs are deterministic. The FM dpcc-r-tight top-left outlier (4.6–4.7s) appears in both seeds.

7. Recommended methods for robust deployment

dpcc-c-tightened ($dt=1.0$) and dpcc-t-tightened are the safest choices: both achieve $\geq 80\%$ G&C; averaged across all 4 seed $\times$ model configurations. post_proc-tightened is also reliable for Diffusion. Avoid dpcc-r (untightened) and baseline methods for constrained scenarios.

Data sources: Seed 5 — diffusion.txt & FM.txt (uploaded). Seed 6 — raw logs (previous analysis). Model A = GaussianDiffusion (diffusion branch). Model B = GaussianDiffusion (flow_matching branch). Stability threshold: $|Seed5-Seed6| \leq 16.7pp$ (i.e. $\leq 1/6$ of scale). All step/violation counts averaged over 2 episodes per seed.